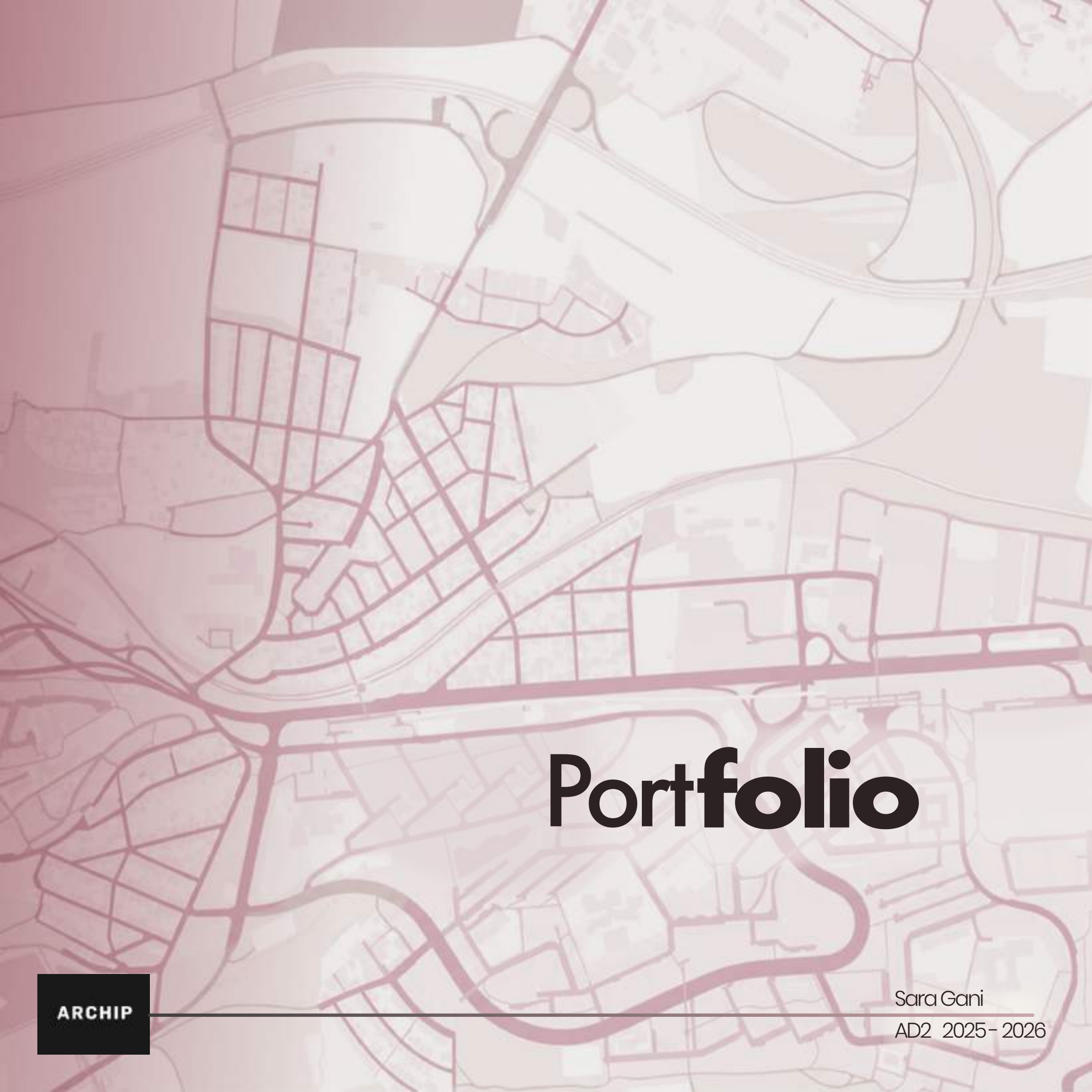




Portfolio

ARCHIP

Sara Gani

AD2 2025 - 2026

Brief	4
Introduction	5-9
Case study	12-15
Proposal	18-25
Surroundings	26-27
EUI,KPI,LCA	28-33
Appendix	34

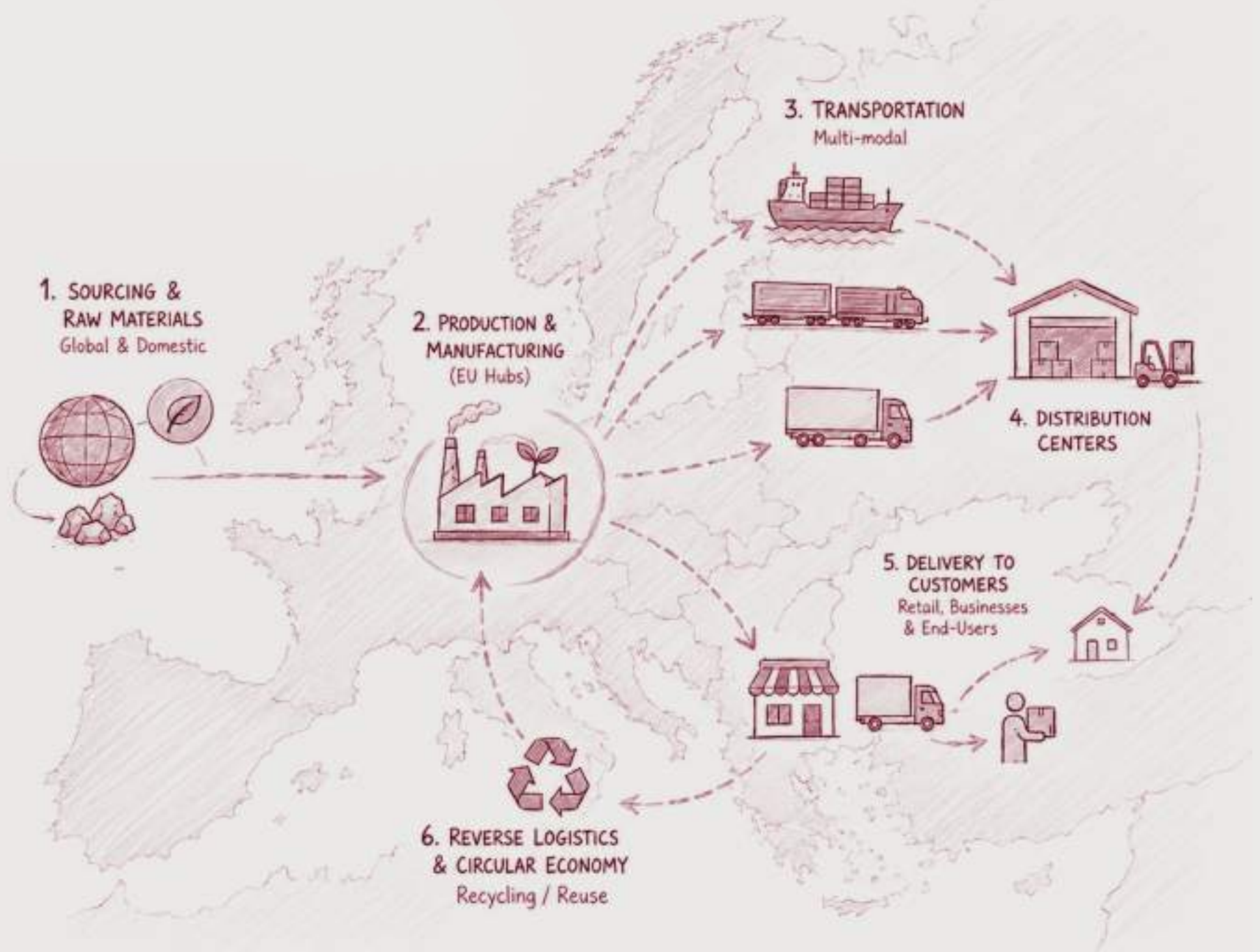
CONTENTS

Brief

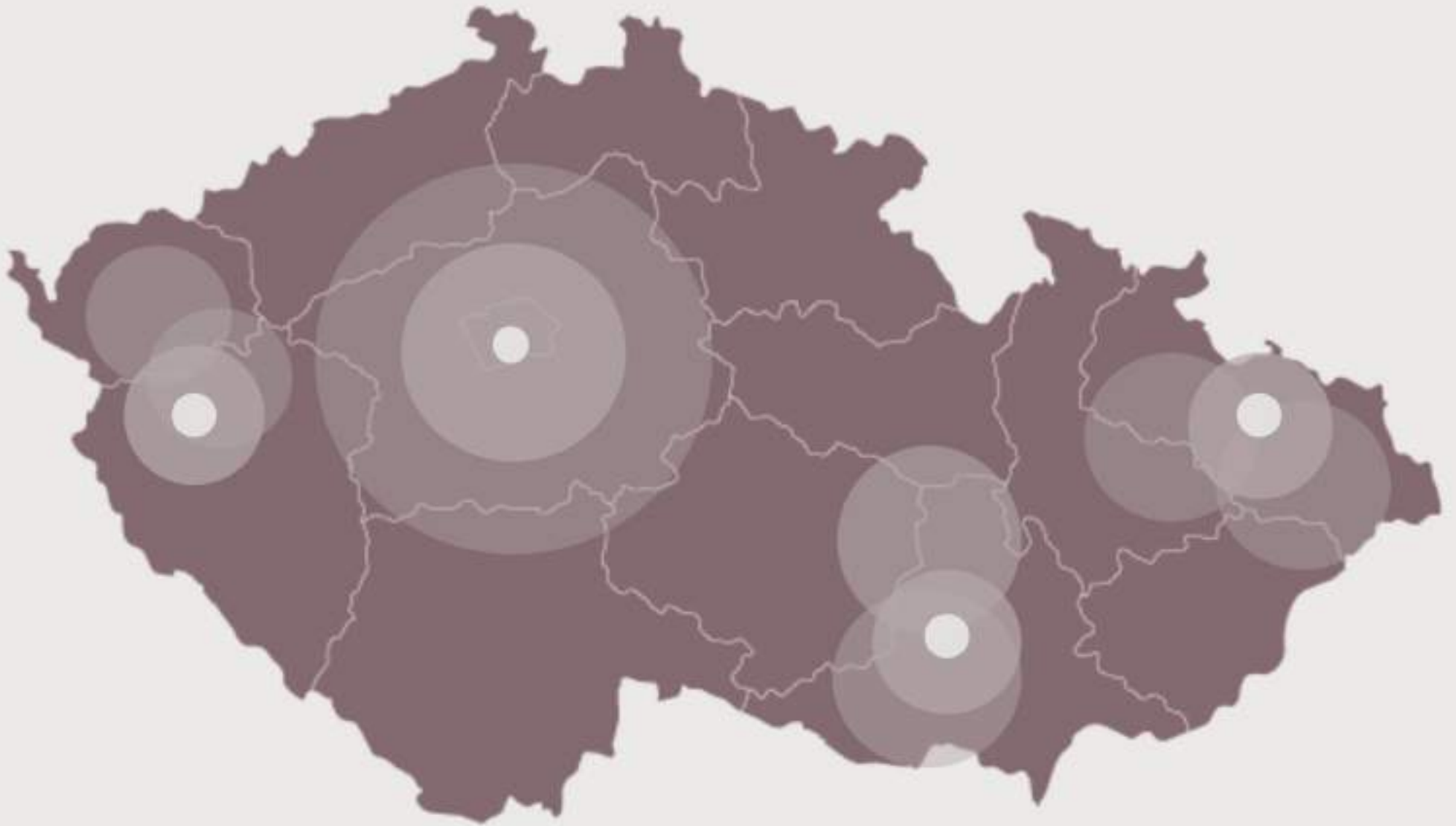
This studio explores how logistics facilities can be redesigned to become sustainable and community focused parts of the city rather than isolated industrial buildings. Working with industry partners including Ramboll, Henning Larsen, and Panattoni, students will use real performance data and environmental analysis to develop innovative logistics strategies that improve energy performance, biodiversity, stormwater management, and urban integration.

The studio combines evidence-based design, collaboration with professionals, and public presentation opportunities, giving students experience in creating realistic and impactful architectural proposals for the future of logistics development.





Czech Republic logistics infrastructure is primarily concentrated around major urban and industrial hubs such as PRAGUE, BRNO, PLZEN, and OSTRAVA due to their strong transport connections, industry, and access to major European trade routes.



Storage and logistics In PRAGUE



In today's consumer-oriented society, these areas are indispensable and their importance continues to grow with the increasing demand for online shopping and delivery of goods and food.



Old warehouses were built for outdated industrial needs, making them inefficient for modern logistics. They often have limited space, outdated infrastructure, higher maintenance costs, and poor operational flexibility.

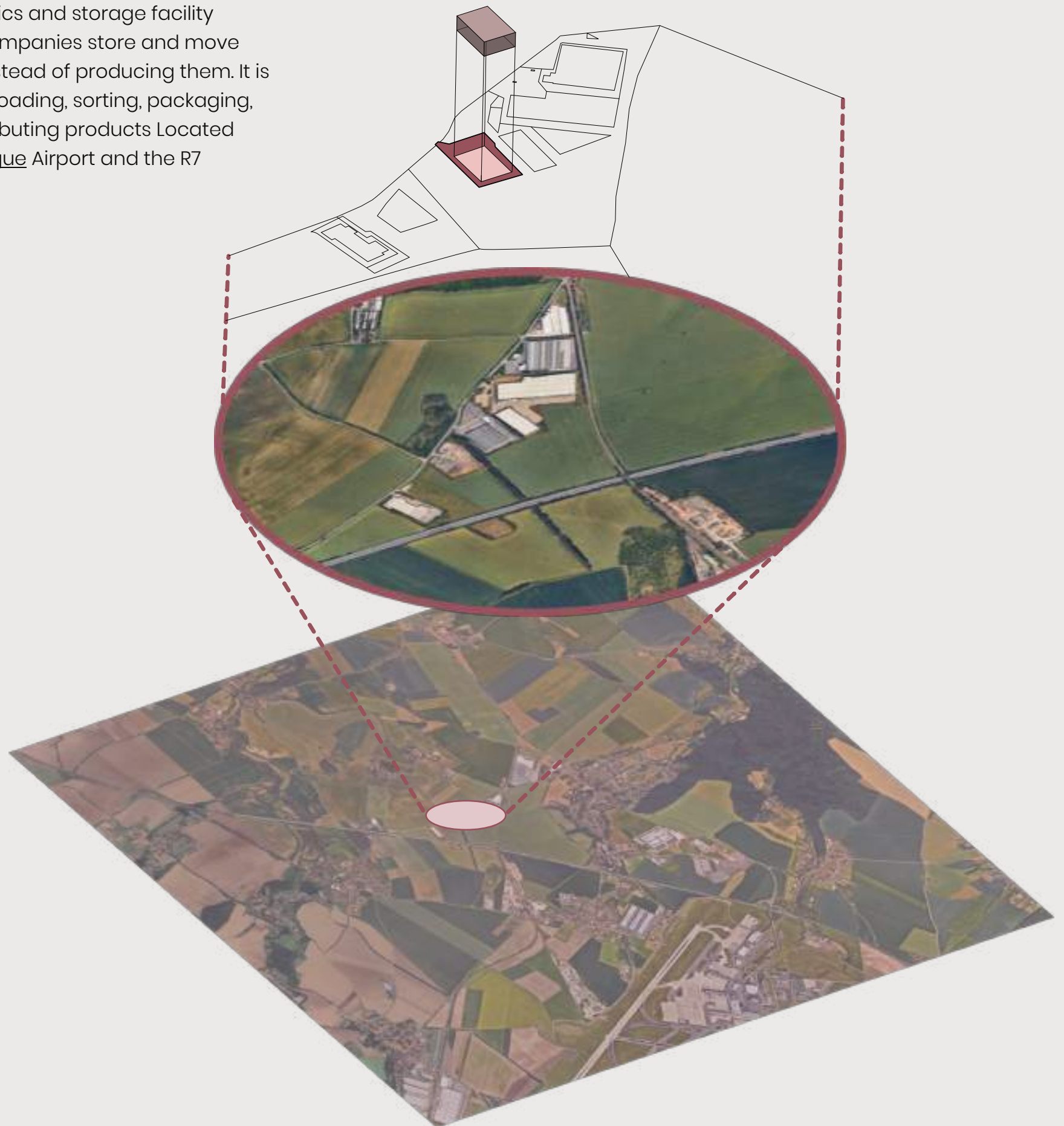
New warehouses are designed for efficiency, scalability, and modern logistics operations. They feature advanced infrastructure, energy-efficient systems, larger storage capacity, and technology-ready spaces that support future growth.



How can
new
strategies
impact this
changing
industry?

Case study

The existing Wakestone Logistics s.r.o. is a logistics and storage facility where companies store and move goods instead of producing them. It is used for loading, sorting, packaging, and distributing products. Located near Prague Airport and the R7 highway.



The existing WAKESTONE LOGISTICS facility covers around 5,500 m² and divided into two halls, with an additional 3,000 m² office area. The warehouse operates through pallet racks, shelf racks, and mostly manual handling systems. It functions mainly as an industrial storage and distribution space for businesses.

Current Problems :

The site faces several problems due to its isolated location, mainly connected only to the airport and highway. Cars and trucks share the same circulation routes, creating unsafe and inefficient movement, while the offices are buried inside the warehouse with limited daylight and connection to public spaces.

PARKING / CAR SPACE

TRUCK CIRUCLATION

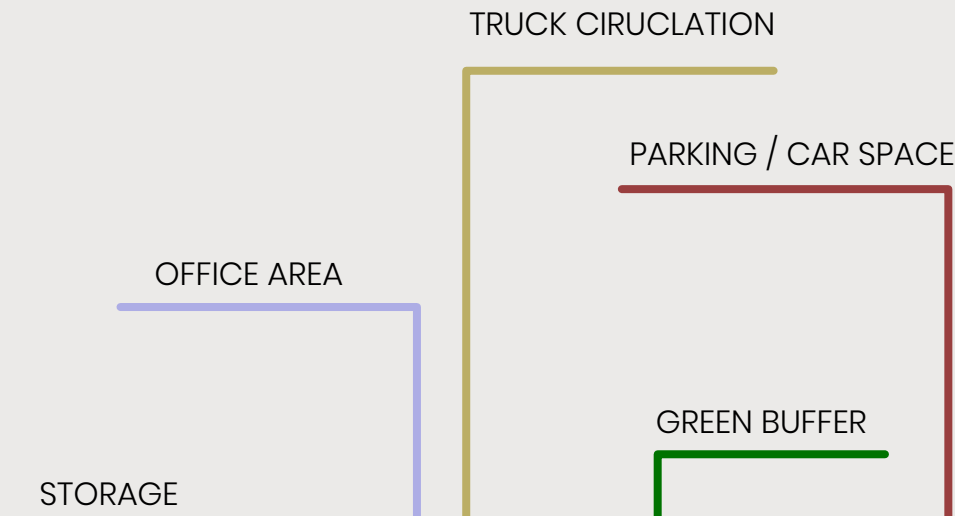
OFFICE/ WORKER COMFORT



The proposal in existing context introduces an L-shaped building that separates human spaces such as offices, control rooms, and a restaurant from logistics areas including high-bay storage, loading docks, and dispatch. It also organises the site with clear circulation routes, truck operations and parking and a green buffer in between logistics operations and human-focused spaces.

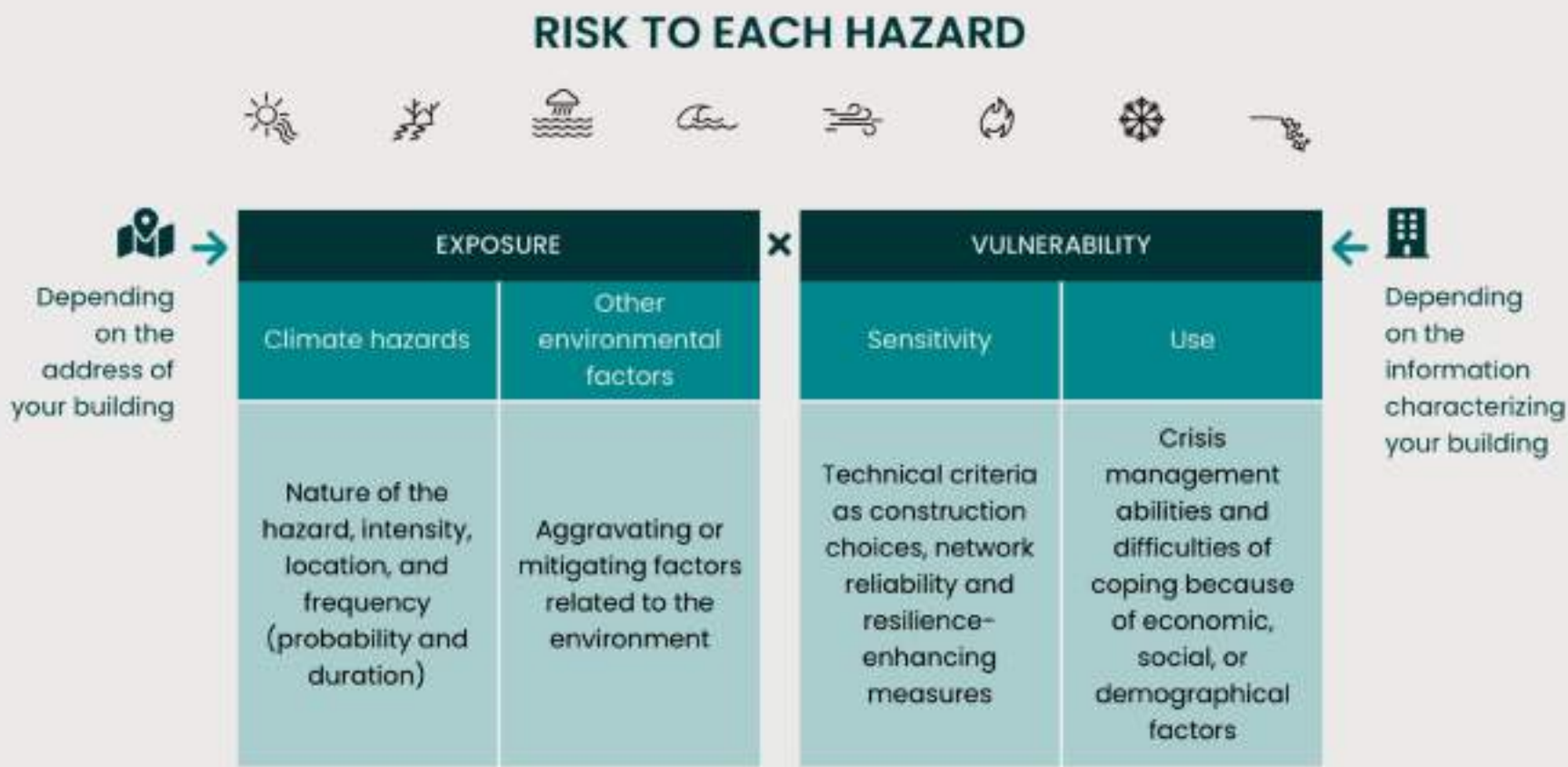
Solutions in existing context:

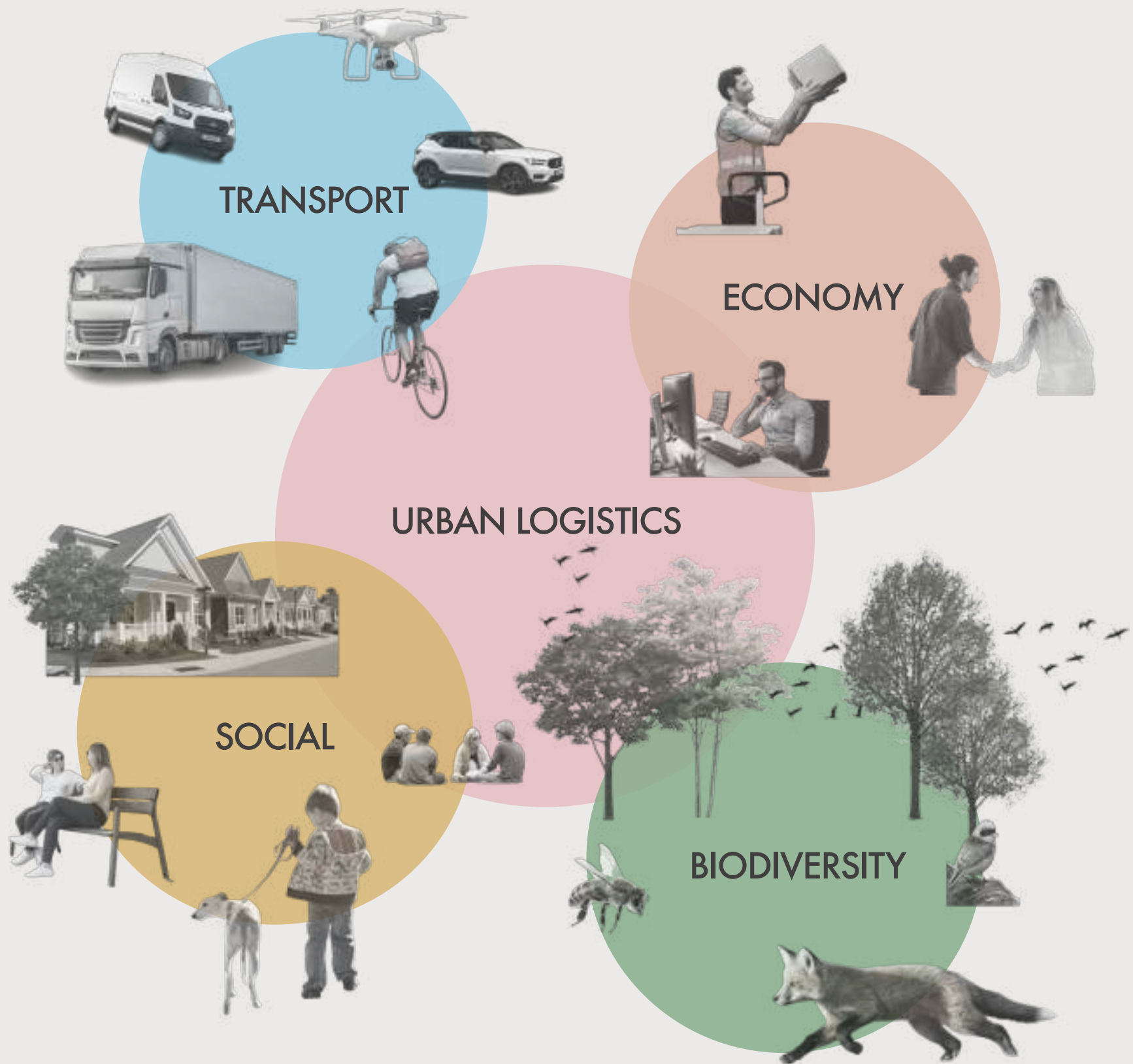
This approach solves the problems of unsafe mixed circulation, inefficient spatial organisation, and poor separation between industrial and working environments. It improves safety, reduces noise and conflict between users, and creates a clearer, more efficient relationship between



Green infrastructure strategies are introduced to reduce environmental impact, improve stormwater management, and increase biodiversity. These interventions also help soften the industrial site and reconnect it with its surrounding landscape.

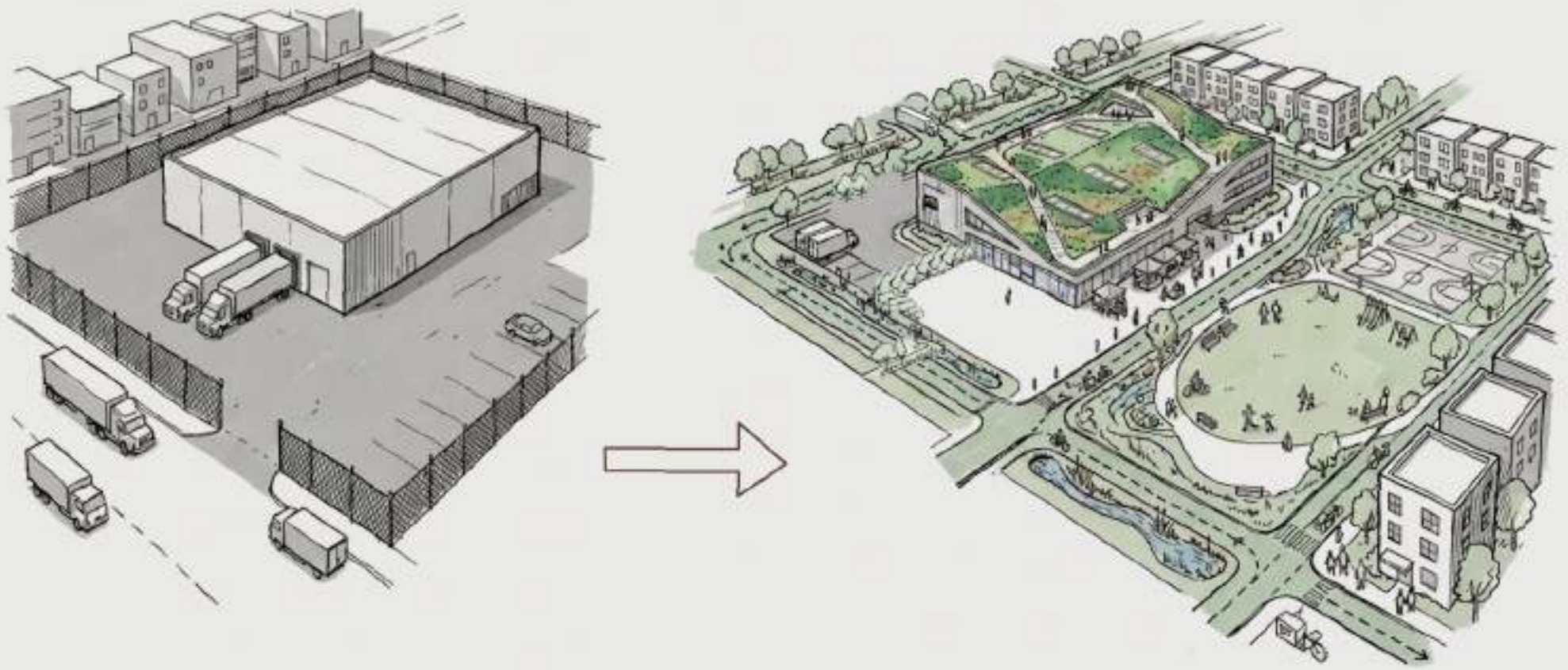
Creating and managing artificial water features	        
Creating a green roof	        
Greening the facade	        
Desealing soil	        
Planting around the building	        
Restoring a watercourse	        





Proposal

Rethinking the relationship between logistics infrastructure and residential communities by transforming an isolated warehouse into a publicly accessible and environmentally integrated hub.





The existing isolated logistics site in Tuchoměřice-Středokluky, is located in a rural, disconnected area surrounded by fields, making it highly dependent on highway access and limiting its connection to surrounding communities.



The new location is within a more urbanised context, Pivoňská, Praha 14 where the logistics function can be better integrated with residential and mixed-use areas. This shift allows the project to improve accessibility, reduce isolation, and create a stronger relationship between industrial activity and the city.

Site analysis

Pivoňská, Praha 14

Buildings



Roads



Green

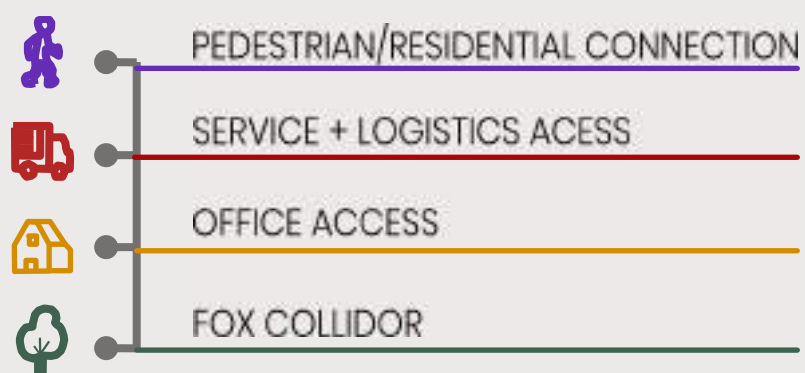
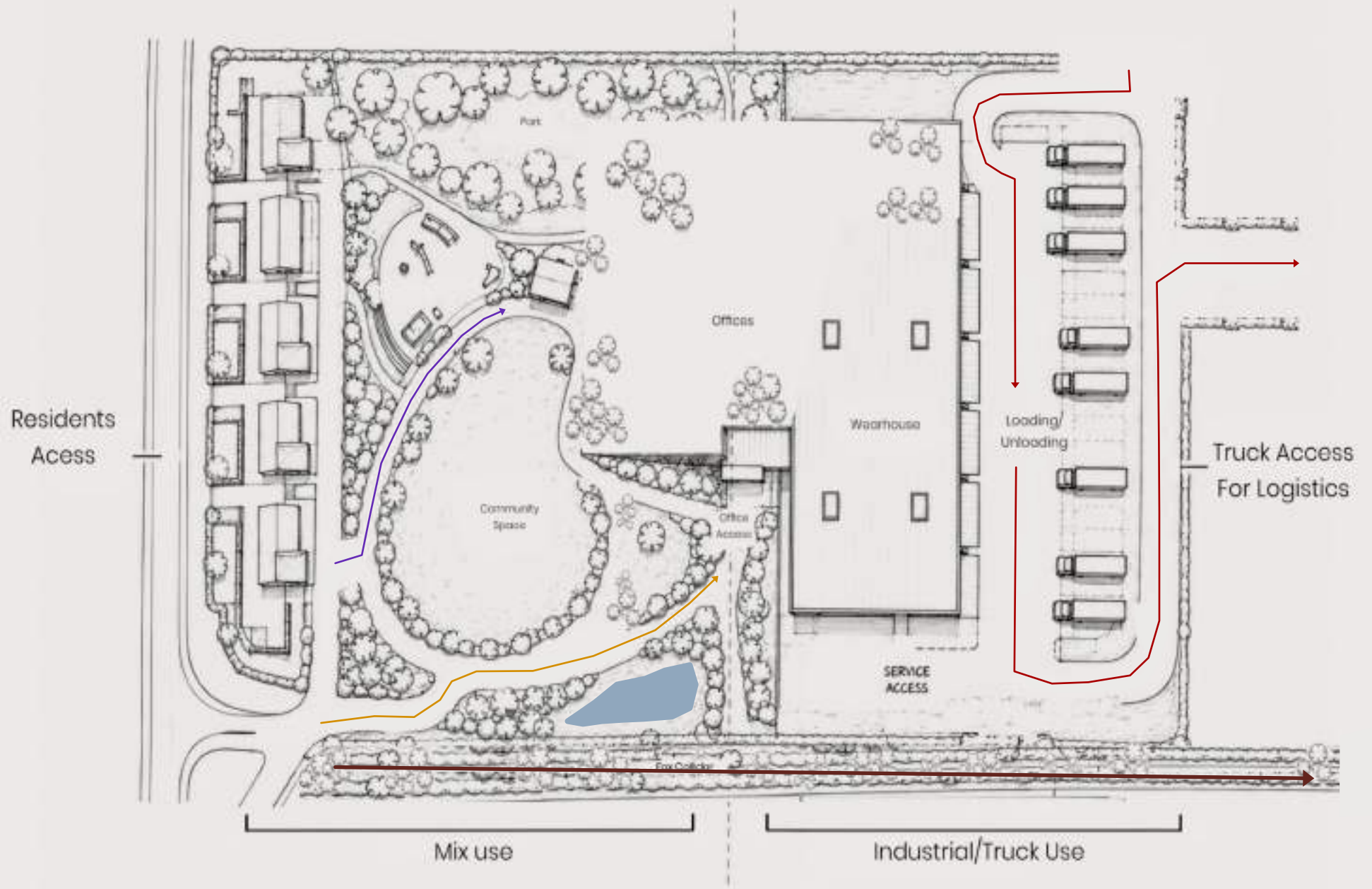


Public



Site

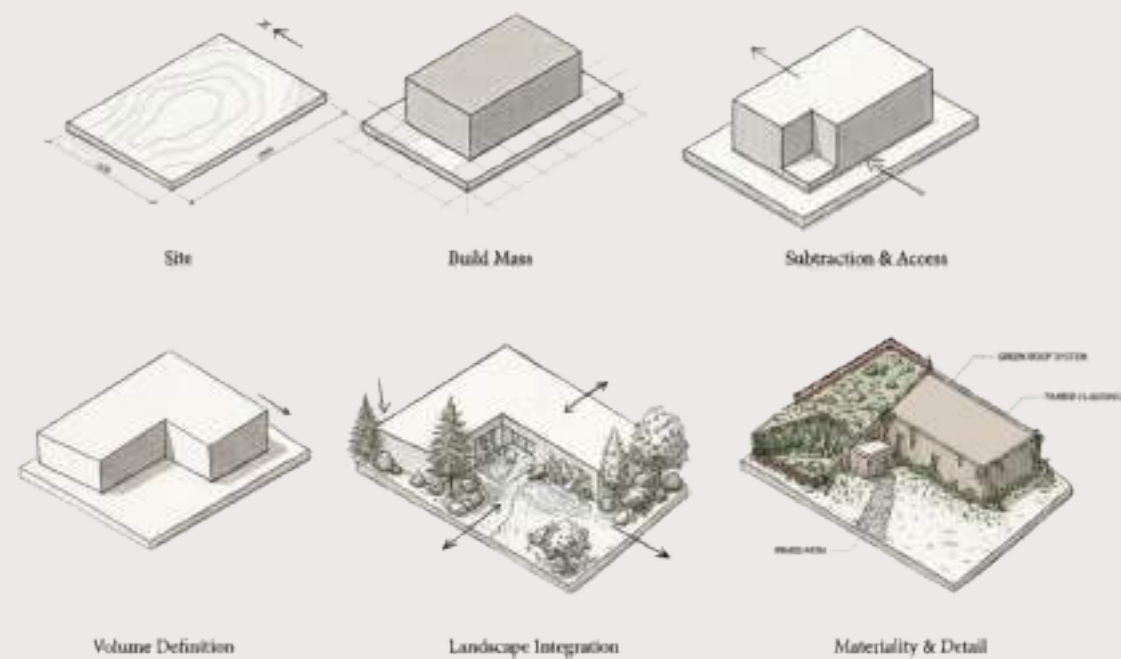






Through landscape, shared public spaces, and ecological strategies, the design creates a balance between operational efficiency and community wellbeing.

The design is informed by an understanding of pedestrian movement and the surrounding environment, using evidence-based analysis to guide spatial and functional decisions.



Infrastructure as Landscape, Where Logistics Meets Life.

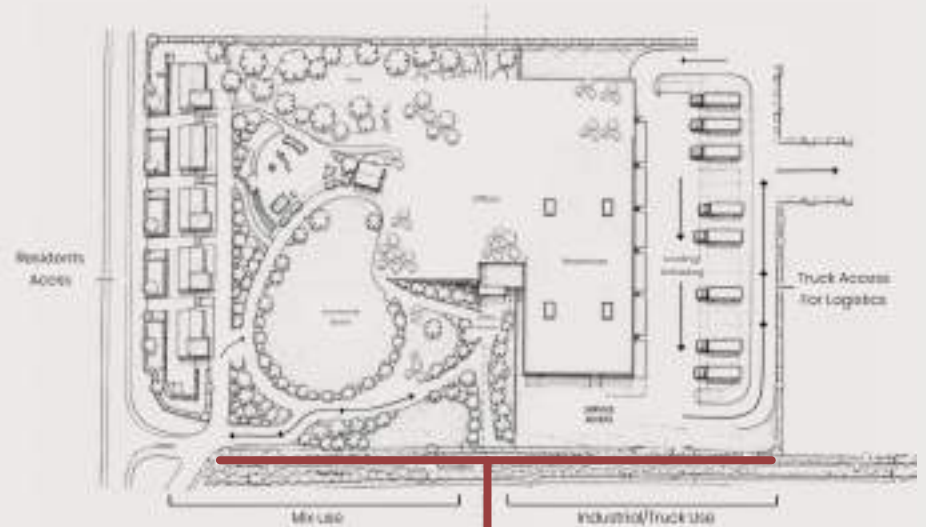
By dissolving the line between industry and community, we transform the warehouse from a silent neighbor into a living landscape.





Surroundings

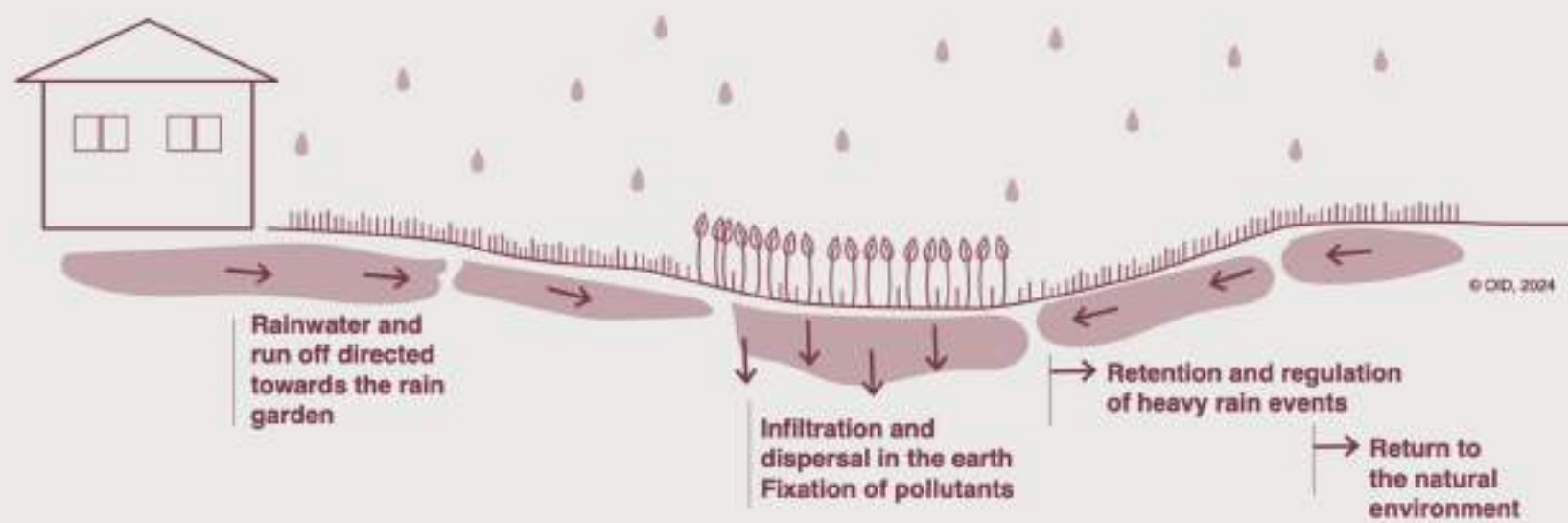
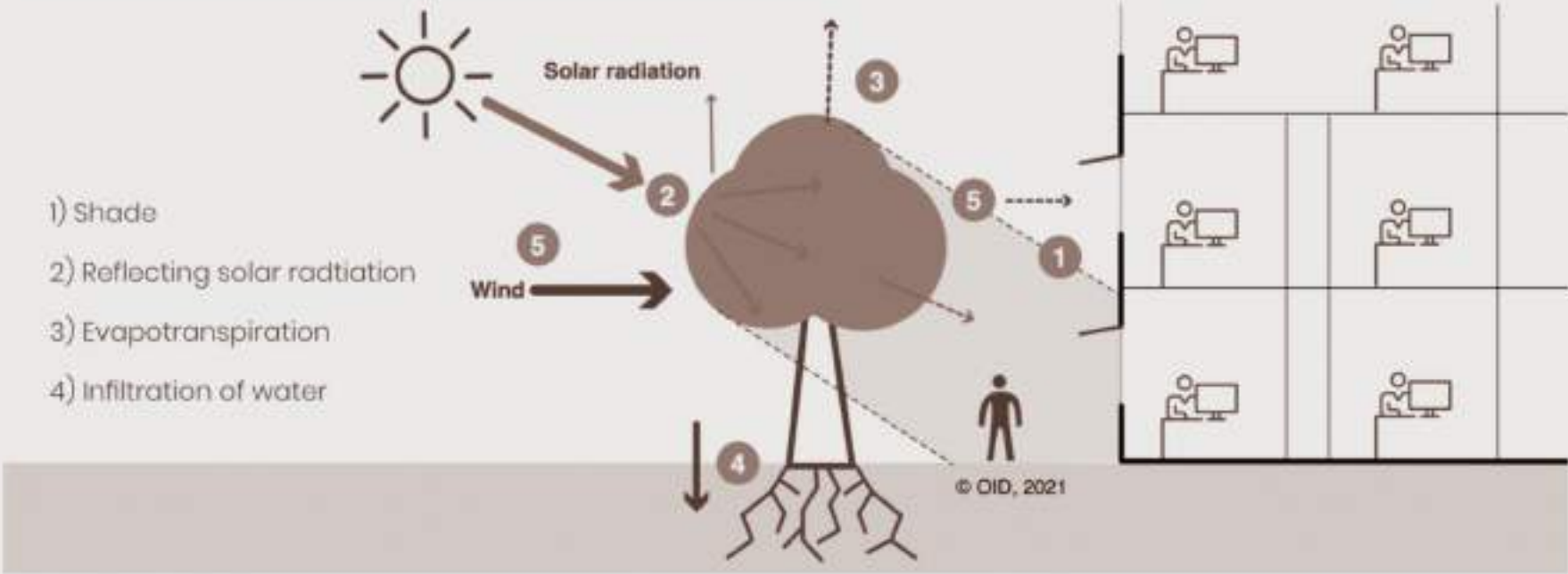
Native plants in the Černý Most area of Prague are characteristic of the Bohemian lowland landscape, consisting of thermo-philic oak forests, oak-hornbeam forests, and steppe grasslands. While the area is heavily developed, the surrounding, less-disturbed, and agricultural zones still contain typical Czech flora.



Creating a dedicated fox corridor to allow safe wildlife movement across the site, reducing the risk of animals crossing railway tracks. This provides a protected ecological route that replaces unsafe movement patterns along the rail line and supports local biodiversity within the industrial landscape.



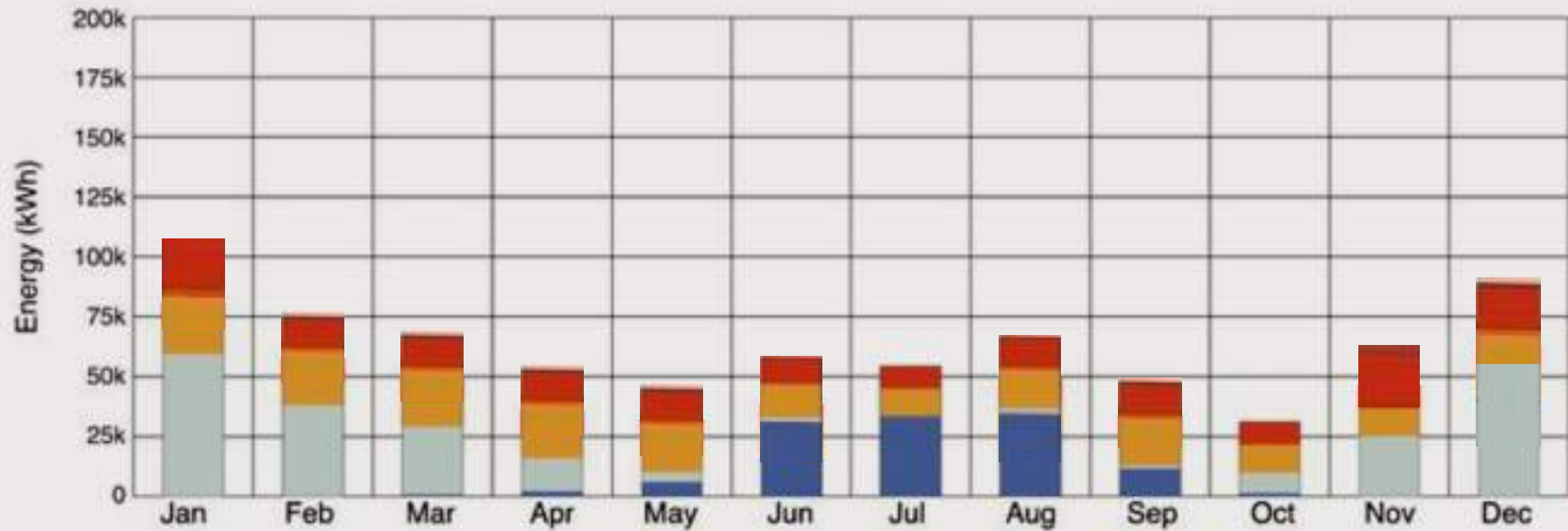
ECOSYSTEM SERVICES PROVIDED BY VEGETATION NEAR A BUILDING



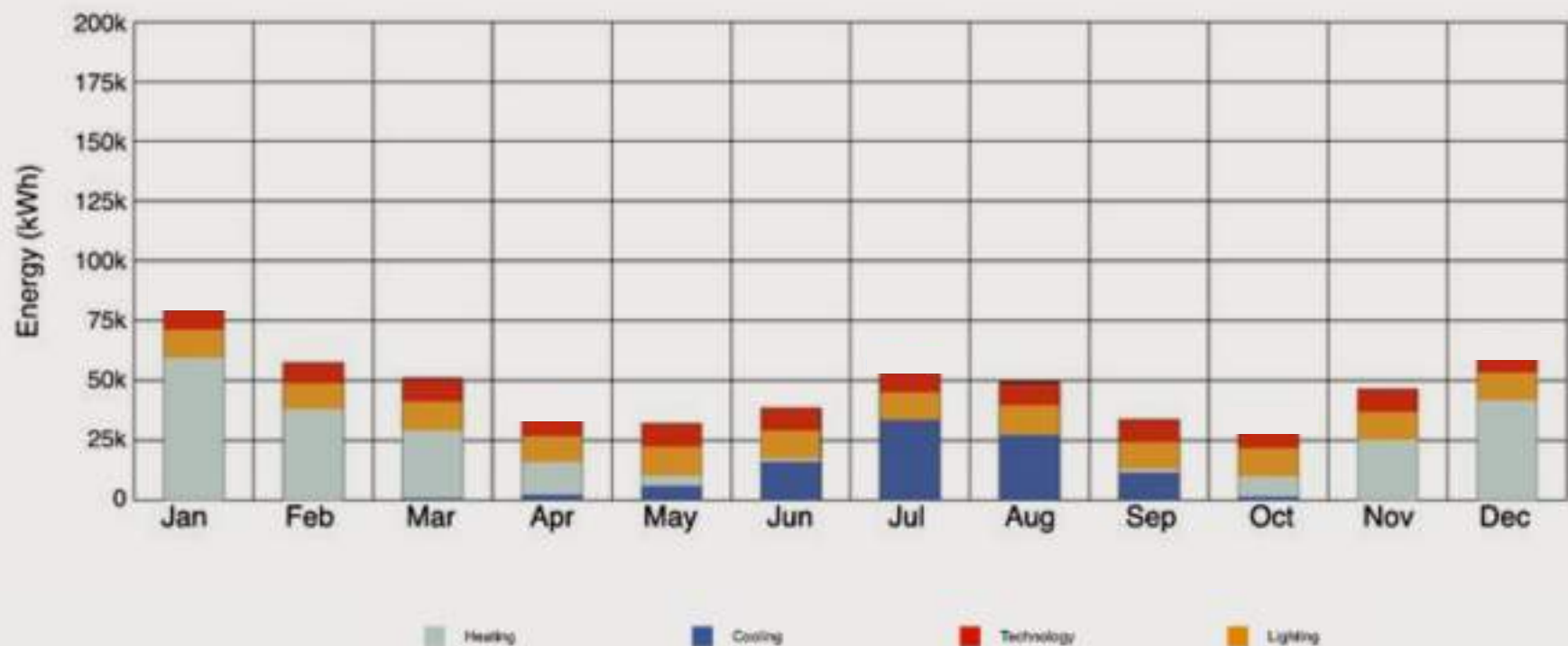
EUI

Energy Use Intensity (EUI) measures the total energy consumed by a building relative to its size, usually expressed in kWh/m²/year. It is a key indicator of building efficiency and operational performance. A lower EUI reflects better energy efficiency and reduced environmental impact.

BEFORE:



AFTER:



KPI

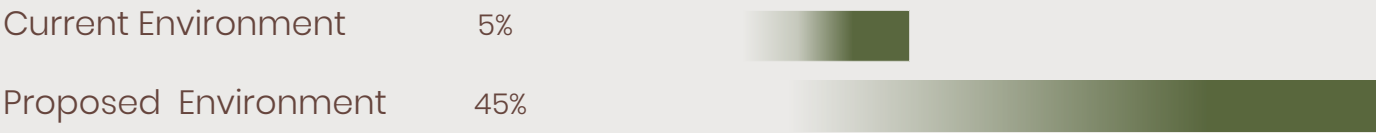
Pedestrain access (m²/year)



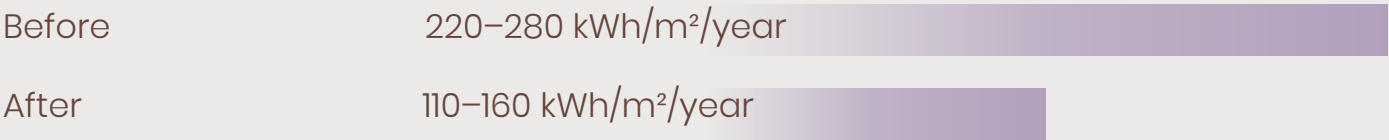
Embodied Carbon (kgCO₂e/kg)



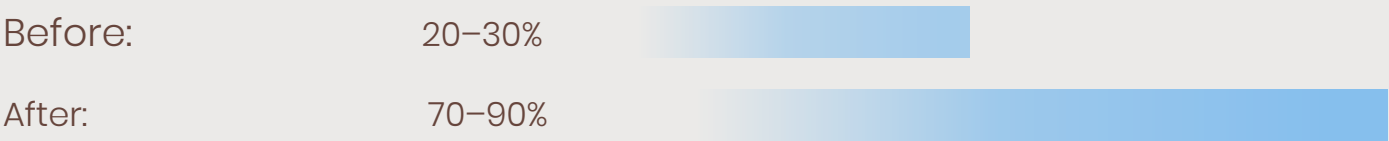
Biodiversity Gain (%)



Energy use per m²



Water managment (% water reuse)



Traffic sepration (%)

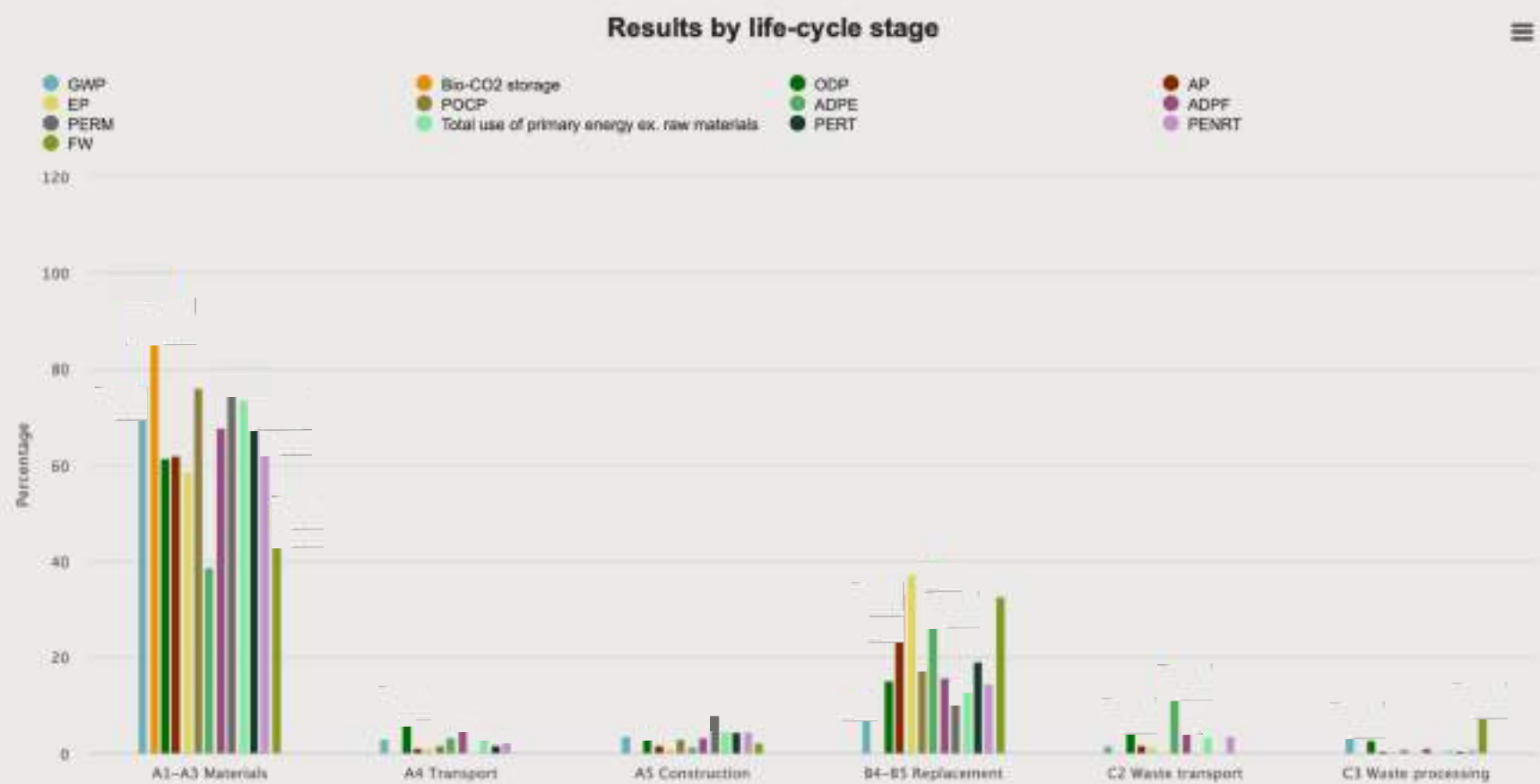


Daylight comfort (%)



LCA

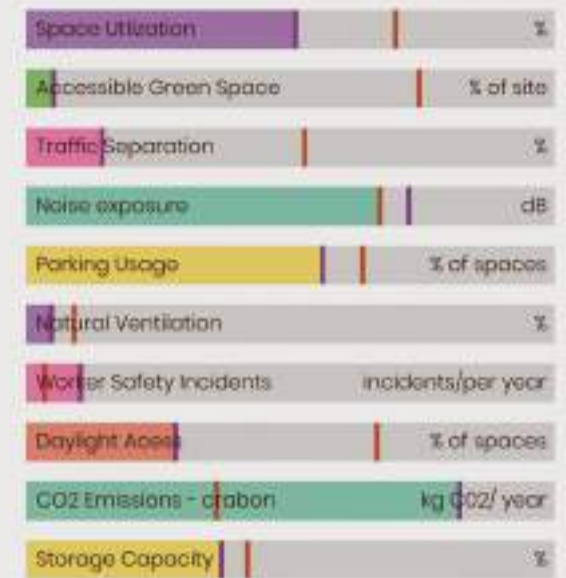
Life Cycle Assessment (LCA) is a method used to evaluate the environmental impact of a building or product across its entire lifespan. It considers all stages, from material extraction and construction to operation, maintenance, and eventual demolition or reuse. By analyzing factors such as carbon emissions, energy use, and waste generation, LCA helps identify more sustainable design choices.



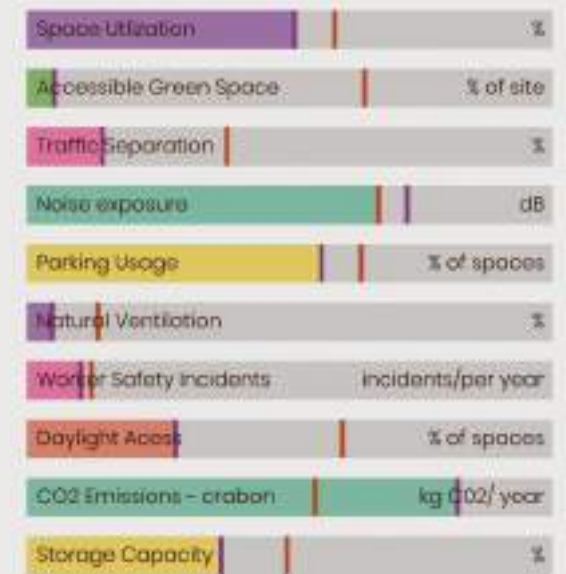
APPENDIX



This layout that separates human spaces such as offices, control rooms, and a café/shop with products that are stored in the logistics area, Circulation routes for trucks, cars, and pedestrians all worked well together and even after exploring other solutions this one worked the best.



The concept separates the program into two buildings: one for logistics and storage, and another for offices, control rooms, and public spaces like a café/shop. Truck, car, and pedestrian circulation are separated even though this didn't work out because of the amount of space being utilized.



The typology changes from one large warehouse into a smaller campus-style layout. Instead of a single industrial box, the program is divided into smaller buildings for logistics, storage, offices, and public spaces. The storage got lower and also wasn't efficient for the trucks.



2026

Saragani@archip.eu

ARCHIP